

A Unit-Window Update Theorem for Density-Based Matroid Decompositions

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Abstract

Huang and Sellier introduced the density-based decomposition of a subset of a matroid in their work on robust sparsification for matroid intersection. They explicitly ask whether, when the subset is changed by one insertion or deletion, the decomposition can be updated more efficiently than recomputing it from scratch. This note gives an exact affected-window update theorem. If an element u is inserted into X and $a = \tilde{\rho}_X(u)$ is its old associated density, then it is enough to recompute the old blocks whose densities lie in the unit interval $[a, a + 1]$, together with u , after contracting the unchanged higher-density prefix. If an element u is deleted from X and $a = \tilde{\rho}_X(u)$ is its old associated density, then it is enough to recompute the old blocks whose densities lie in $[a - 1, a]$, after removing u and contracting the unchanged higher-density prefix.

This is genuine partial progress on the update question, not a full worst-case dynamic data structure. The affected unit window can contain all of X , in which case the method degenerates to static recomputation. The contribution is therefore an exact output-sensitive reduction from a one-element update to one static decomposition on the affected density window.

1 Introduction and status of the result

Let $M = (V, \mathcal{I})$ be a finite loopless matroid. For $X \subseteq V$, the density-based decomposition of X is the greedy decomposition obtained by repeatedly taking a maximum-cardinality densest set in the current contraction. Huang and Sellier use this decomposition to define associated densities, which play the role of degrees in their density-constrained subset sparsifier for matroid intersection [1]. They also ask whether this decomposition can be updated efficiently when the maintained subset is changed by one insertion or deletion, rather than recomputed from scratch.

This note proves the following result. The statement is intentionally phrased as partial progress.

Theorem 1.1 (informal affected-window theorem). *Assume a current density-based decomposition of X is stored. If $u \notin X$ is inserted and $a = \tilde{\rho}_X(u)$, then the exact new decomposition of $X \cup \{u\}$ is obtained by leaving all old blocks of density greater than $a + 1$ and less than a unchanged, and recomputing only the old blocks of density in $[a, a + 1]$, together with u , in the minor obtained after contracting the unchanged higher-density prefix.*

If $u \in X$ is deleted and $a = \tilde{\rho}_X(u)$, then the exact new decomposition of $X \setminus \{u\}$ is obtained by leaving all old blocks of density greater than a and less than $a - 1$ unchanged, and recomputing only the old blocks of density in $[a - 1, a]$, with u removed, in the minor obtained after contracting the unchanged higher-density prefix.

The theorem is an exact locality result in density space. It is not a worst-case sublinear update-time theorem: the interval $[a, a + 1]$ or $[a - 1, a]$ may contain every old block. Nevertheless,

whenever the affected window is a proper subinstance, the update is reduced to a strictly smaller static decomposition problem.

2 Density-based decompositions

For a matroid minor N and a set A in its ground set, define

$$\rho_N(A) = \begin{cases} |A|/r_N(A), & A \neq \emptyset \text{ and } r_N(A) > 0, \\ 0, & A = \emptyset, \\ +\infty, & A \neq \emptyset \text{ and } r_N(A) = 0. \end{cases}$$

Only finite-density nonempty blocks will occur below. The convention for zero-rank nonempty sets is included only to make the densest-set language unambiguous.

Definition 2.1 (density-based decomposition). Let $X \subseteq V$. Set $P_0 = \emptyset$. Having chosen P_{i-1} , if $P_{i-1} \neq X$, choose $U_i \subseteq X \setminus P_{i-1}$ to be the maximum-cardinality member among all subsets of $X \setminus P_{i-1}$ maximizing density in the minor $(M/P_{i-1})|_{X \setminus P_{i-1}}$. Set $P_i = P_{i-1} \cup U_i$ and continue until $P_s = X$.

The ordered list

$$\mathcal{D}_X = (U_1, \dots, U_s)$$

of nonempty blocks is the density-based decomposition of X . Its block densities are

$$\delta_i = \rho_{M/P_{i-1}}(U_i) = \frac{|U_i|}{r_M(P_i) - r_M(P_{i-1})}.$$

The maximum-cardinality densest set is unique. Indeed, if λ is the maximum density and A, B are densest, then the function

$$f(S) = |S| - \lambda r(S)$$

is supermodular and satisfies $f(S) \leq 0$ for every S , while $f(A) = f(B) = 0$. Hence $f(A \cup B) = f(A \cap B) = 0$, so $A \cup B$ is also densest. Thus two maximum-cardinality densest sets must have the same union and therefore are equal.

Lemma 2.2 (basic properties). *For the decomposition $\mathcal{D}_X = (U_1, \dots, U_s)$:*

- (i) *each prefix $P_i = U_1 \cup \dots \cup U_i$ is closed in the restriction $M|X$;*
- (ii) *all nonempty block densities are finite and satisfy*

$$\delta_1 > \delta_2 > \dots > \delta_s > 0.$$

Proof. We prove the closure assertion by induction. The empty set is closed in $M|X$ because M is loopless. Suppose P_{i-1} is closed in $M|X$. Then no element of $X \setminus P_{i-1}$ is a loop in the contracted restriction $(M/P_{i-1})|_{X \setminus P_{i-1}}$, so every nonempty remaining set has positive rank. Let $e \in X \setminus P_i$. If e were in the closure of U_i in $(M/P_{i-1})|_{X \setminus P_{i-1}}$, then $U_i \cup \{e\}$ would have the same rank as U_i and larger cardinality. This would give strictly larger density, contradicting the choice of U_i . Hence U_i is closed in the current restriction, equivalently P_i is closed in $M|X$.

The finiteness and positivity of the densities follow from the preceding paragraph. To prove strict decrease, work in the minor M/P_{i-1} . Let $r_i = r_{M/P_{i-1}}(U_i)$ and let $r_{i+1} = r_{M/P_i}(U_{i+1})$. If $\delta_{i+1} > \delta_i$, then the density of $U_i \cup U_{i+1}$ in M/P_{i-1} is the weighted average

$$\frac{|U_i| + |U_{i+1}|}{r_i + r_{i+1}} = \frac{\delta_i r_i + \delta_{i+1} r_{i+1}}{r_i + r_{i+1}},$$

which is greater than δ_i , contradicting that U_i was densest. If $\delta_{i+1} = \delta_i$, then $U_i \cup U_{i+1}$ has the same density as U_i and strictly larger cardinality, contradicting the maximum-cardinality tie-breaking. Therefore $\delta_i > \delta_{i+1}$. $\square$

Definition 2.3 (associated density). Let $\mathcal{D}_X = (U_1, \dots, U_s)$ and $P_i = U_1 \cup \dots \cup U_i$. For $e \in V$, define the associated density of e with respect to X by

$$\tilde{\rho}_X(e) = \begin{cases} \delta_i, & i = \min\{j : e \in \text{cl}_M(P_j)\}, \text{ if } e \in \text{cl}_M(X), \\ 0, & e \notin \text{cl}_M(X). \end{cases}$$

By Lemma 2.2, if $e \in U_i$ then i is the least index with $e \in \text{cl}_M(P_i)$, so $\tilde{\rho}_X(e) = \delta_i$.

3 Huang–Sellier one-element sensitivity lemmas

We use the following consequences of Lemmas 20 and 21 of Huang and Sellier [1]. They are stated here in the form needed for the update theorem.

Lemma 3.1 (insertion sensitivity, Huang–Sellier). *Let $Y = X \cup \{u\}$ with $u \notin X$, and put $a = \tilde{\rho}_X(u)$. Then:*

- (i) $\tilde{\rho}_Y(v) \geq \tilde{\rho}_X(v)$ for every $v \in V$;
- (ii) $a \leq \tilde{\rho}_Y(u) \leq a + 1$;
- (iii) if $v \in X$ and either $\tilde{\rho}_X(v) < a$ or $\tilde{\rho}_X(v) > a + 1$, then $\tilde{\rho}_Y(v) = \tilde{\rho}_X(v)$.

Lemma 3.2 (deletion sensitivity, Huang–Sellier). *Let $Y = X \setminus \{u\}$ with $u \in X$, and put $a = \tilde{\rho}_X(u)$. Then:*

- (i) $\tilde{\rho}_Y(v) \leq \tilde{\rho}_X(v)$ for every $v \in V$;
- (ii) $a - 1 \leq \tilde{\rho}_Y(u) \leq a$;
- (iii) if $v \in Y$ and either $\tilde{\rho}_X(v) > a$ or $\tilde{\rho}_X(v) < a - 1$, then $\tilde{\rho}_Y(v) = \tilde{\rho}_X(v)$.

4 The unit-window lemma

The next lemma is the main additional observation. Huang–Sellier’s lemmas immediately show that elements outside the relevant unit interval do not change. Applying the opposite update in reverse also shows that elements inside the relevant interval cannot move outside it.

Lemma 4.1 (two-sided unit-window lemma). *Let $\mathcal{D}_X = (U_1, \dots, U_s)$ have densities $\delta_1 > \dots > \delta_s$.*

- (a) **Insertion.** *Let $Y = X \cup \{u\}$ with $u \notin X$, and let $a = \tilde{\rho}_X(u)$. Then $\tilde{\rho}_Y(u) \in [a, a + 1]$. Moreover, for every $v \in X$:*

$$\tilde{\rho}_X(v) \in [a, a + 1] \implies \tilde{\rho}_Y(v) \in [a, a + 1],$$

and

$$\tilde{\rho}_X(v) \notin [a, a + 1] \implies \tilde{\rho}_Y(v) = \tilde{\rho}_X(v).$$

(b) **Deletion.** Let $Y = X \setminus \{u\}$ with $u \in X$, and let $a = \tilde{\rho}_X(u)$. Then, for every $v \in Y$:

$$\tilde{\rho}_X(v) \in [a-1, a] \implies \tilde{\rho}_Y(v) \in [a-1, a],$$

and

$$\tilde{\rho}_X(v) \notin [a-1, a] \implies \tilde{\rho}_Y(v) = \tilde{\rho}_X(v).$$

Consequently, in the insertion case every changed old element and the inserted element have new associated density in $[a, a+1]$; in the deletion case every changed remaining element has new associated density in $[a-1, a]$.

Proof. For insertion, Lemma 3.1 gives $\tilde{\rho}_Y(u) \in [a, a+1]$ and also gives equality for all old elements with old associated density outside $[a, a+1]$. It remains to show that an old element whose old density lies in $[a, a+1]$ cannot move above $a+1$. Let $v \in X$, put

$$d = \tilde{\rho}_X(v), \quad D = \tilde{\rho}_Y(v), \quad b = \tilde{\rho}_Y(u),$$

and suppose $d \in [a, a+1]$. Monotonicity gives $D \geq d \geq a$. If $D > a+1$, then $D > b$, since $b \leq a+1$. Now delete u from Y to recover X . Applying Lemma 3.2 to this deletion, with pre-deletion associated density of u equal to b , shows that every element whose pre-deletion associated density is greater than b is unchanged by deleting u . Hence $d = D$, contradicting $d \leq a+1 < D$. Therefore $D \leq a+1$, and so $D \in [a, a+1]$.

For deletion, Lemma 3.2 gives equality for every remaining element with old associated density outside $[a-1, a]$. Let $v \in Y$, put

$$d = \tilde{\rho}_X(v), \quad D = \tilde{\rho}_Y(v), \quad b = \tilde{\rho}_Y(u),$$

and suppose $d \in [a-1, a]$. Monotonicity gives $D \leq d \leq a$. If $D < a-1$, then $D < b$, since $b \geq a-1$. Now insert u back into Y to recover X . Applying Lemma 3.1 to this insertion, with pre-insertion associated density of u equal to b , shows that every old element whose pre-insertion associated density is less than b is unchanged by inserting u . Hence $d = D$, contradicting $D < a-1 \leq d$. Therefore $D \geq a-1$, and so $D \in [a-1, a]$. $\square$

Corollary 4.2 (unchanged blocks outside the unit window). *In the insertion setting of Lemma 4.1, every old block with density $> a+1$ or $< a$ is a block of the new decomposition, with the same elements and the same density. In the deletion setting, every old block with density $> a$ or $< a-1$ and not containing the deleted element is a block of the new decomposition, with the same elements and the same density.*

Proof. We prove the insertion case; the deletion case is identical. Let U_i be an old block with density $\delta_i \notin [a, a+1]$. Lemma 4.1 says that every $v \in U_i$ still has associated density δ_i . Since new nonempty block densities are strictly decreasing by Lemma 2.2, all elements of the updated set with associated density δ_i lie in one new block. No element from the old unit window or the inserted element can join that block, because their new associated densities lie in $[a, a+1]$, while $\delta_i \notin [a, a+1]$. No element from a different old block outside the window can join it, because distinct old blocks have distinct densities and those densities are unchanged. Thus the new block at density δ_i is exactly U_i . $\square$

5 Local recomputation inside a contracted minor

For disjoint sets $L, A \subseteq V$, write

$$\text{Dec}(L, A)$$

for the density-based decomposition of A in the minor $(M/L)|_A$. Its rank oracle is simulated by the original rank oracle via

$$r_{M/L}(B) = r_M(L \cup B) - r_M(L), \quad B \subseteq A.$$

Lemma 5.1 (contiguous subsequences are local decompositions). *Let $\mathcal{D}_X = (U_1, \dots, U_s)$ and $P_i = U_1 \cup \dots \cup U_i$. If $1 \leq p \leq q \leq s$, then*

$$(U_p, U_{p+1}, \dots, U_q)$$

is the density-based decomposition of $P_q \setminus P_{p-1}$ in the minor $(M/P_{p-1})|_{P_q \setminus P_{p-1}}$.

Proof. After the full greedy procedure has contracted P_{p-1} , it chooses U_p as the maximum-cardinality densest set among all remaining elements $X \setminus P_{p-1}$. Since $P_q \setminus P_{p-1}$ is a subset of that remaining ground set and contains U_p , no subset of $P_q \setminus P_{p-1}$ has density larger than U_p , and no subset with the same density has larger cardinality. Thus U_p is the first block of the local decomposition. Contract U_p and repeat the same argument for $U_{p+1}, \dots, U_q$. $\square$

6 The exact affected-window update algorithm

Let the current decomposition be

$$\mathcal{D}_X = (U_1, \dots, U_s), \quad \delta_1 > \dots > \delta_s.$$

Empty intervals below are allowed; replacing an empty interval means inserting the locally recomputed list at the indicated boundary.

6.1 Insertion

Let $u \notin X$ and set $a = \tilde{\rho}_X(u)$. Define

$$H^+ = \{i : \delta_i > a + 1\}, \quad W^+ = \{i : a \leq \delta_i \leq a + 1\}, \quad T^+ = \{i : \delta_i < a\}.$$

Because the densities are strictly decreasing, these are respectively a prefix, a contiguous interval, and a suffix of the block order. Put

$$L^+ = \bigcup_{i \in H^+} U_i, \quad A^+ = \{u\} \cup \bigcup_{i \in W^+} U_i.$$

The insertion update outputs the concatenation

$$(U_i)_{i \in H^+} \parallel \text{Dec}(L^+, A^+) \parallel (U_i)_{i \in T^+},$$

where the old blocks in each displayed subsequence are kept in their old order.

6.2 Deletion

Let $u \in X$ and set $a = \tilde{\rho}_X(u)$, which is the density of the old block containing u . Define

$$H^- = \{i : \delta_i > a\}, \quad W^- = \{i : a - 1 \leq \delta_i \leq a\}, \quad T^- = \{i : \delta_i < a - 1\}.$$

Again these form a prefix, a contiguous interval, and a suffix. Put

$$L^- = \bigcup_{i \in H^-} U_i, \quad A^- = \left(\bigcup_{i \in W^-} U_i \right) \setminus \{u\}.$$

The deletion update outputs the concatenation

$$(U_i)_{i \in H^-} \parallel \text{Dec}(L^-, A^-) \parallel (U_i)_{i \in T^-}.$$

If $A^- = \emptyset$, then $\text{Dec}(L^-, A^-)$ is the empty list.

Theorem 6.1 (exact affected-window update). *The insertion and deletion procedures above output exactly the density-based decomposition of the updated set.*

Proof. We first prove insertion. Let $Y = X \cup \{u\}$. By Corollary 4.2, every old block with density greater than $a + 1$ remains a block of the new decomposition, and every old block with density less than a remains a block of the new decomposition. Lemma 4.1 also says that every element in A^+ has new associated density in $[a, a + 1]$. Since new block densities are strictly decreasing, the old blocks with density greater than $a + 1$ must appear before all blocks meeting A^+ , and the old blocks with density less than a must appear after all blocks meeting A^+ . Hence the true new decomposition has the form

$$(U_i)_{i \in H^+} \parallel \mathcal{W} \parallel (U_i)_{i \in T^+},$$

where $\mathcal{W}$ is a contiguous middle subsequence whose ground set is exactly A^+ .

After the unchanged prefix L^+ has been contracted, Lemma 5.1, applied to the true decomposition of Y , implies that this middle subsequence $\mathcal{W}$ is precisely the density-based decomposition of A^+ in $(M/L^+)|_{A^+}$, namely $\text{Dec}(L^+, A^+)$. This is exactly the list produced by the insertion procedure.

The deletion case is the same argument with the interval $[a - 1, a]$. Let $Y = X \setminus \{u\}$. Corollary 4.2 leaves unchanged all blocks with density greater than a and less than $a - 1$. Lemma 4.1 puts every changed remaining element in the density interval $[a - 1, a]$. Therefore the true new decomposition has the form

$$(U_i)_{i \in H^-} \parallel \mathcal{W} \parallel (U_i)_{i \in T^-},$$

where the middle ground set is exactly A^- . Contracting L^- and applying Lemma 5.1 to the true new decomposition shows that $\mathcal{W} = \text{Dec}(L^-, A^-)$. $\square$

7 Oracle complexity and limitation

Let $T_{\text{dec}}(m, h)$ denote the rank-oracle complexity of a chosen static density-decomposition routine on an active set of size m and rank h . For example, using the standard principal-partition approach based on submodular function minimization gives a bound of the form

$$T_{\text{dec}}(m, h) = O(h \log(mh) T_{\text{SFM}}(m)),$$

where $T_{\text{SFM}}(m)$ is the cost of minimizing a submodular function on an m -element ground set. The update theorem is independent of the specific static routine used.

Suppose the current decomposition has s nonempty blocks. For an insertion, the value $a = \tilde{\rho}_X(u)$ can be found by binary search on the prefix chain: the predicate

$$u \in \text{cl}_M(P_i)$$

is monotone in i and is tested by the rank comparison

$$r_M(P_i \cup \{u\}) = r_M(P_i).$$

Thus $O(\log s)$ rank queries find the first prefix spanning u , or certify that no prefix spans u and hence $a = 0$. Once a is known, the density interval $[a, a + 1]$ is found by predecessor/successor search among the stored decreasing densities. The only nontrivial recomputation is on

$$m^+ = |A^+|, \quad h^+ = r_M(L^+ \cup A^+) - r_M(L^+).$$

Thus the update uses

$$O(\log s) + T_{\text{dec}}(m^+, h^+) + O(m^+)$$

rank-oracle work and bookkeeping, up to the representation cost of the maintained block tree.

For a deletion, a is stored as the density of the block containing u . The affected interval is found among the stored densities, and the static recomputation is on

$$m^- = |A^-|, \quad h^- = r_M(L^- \cup A^-) - r_M(L^-).$$

The corresponding bound is

$$O(\log s) + T_{\text{dec}}(m^-, h^-) + O(m^-).$$

Remark 7.1 (why this is partial progress). The theorem gives an exact output-sensitive update bound, but it does not give a worst-case improvement over recomputation from scratch. The affected unit window may contain all old blocks. In that case m^+ or m^- is $\Theta(|X|)$, and the algorithm is essentially a static recomputation. Therefore this note should be read as solving the affected-window localization problem and as partial progress toward the broader dynamic-update question raised by Huang and Sellier.

References

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